Inference I: t-tests, F-tests, and Confidence Intervals

Applied Linear Models I • Lecture 4 of 16 • 90-minute version

Applied Linear Models I

Under normal errors $\varepsilon \sim N(0, \sigma^2 \mathrm{I})$ the procedures are *exact*; for large n they are approximately valid (CLT, Lecture 3).

Min	Part	Key question
12	0. Notation, assumptions, running example	What does every symbol mean?
10	1. Sampling distributions	Where do t and F come from?
15	2. The t-test for one coefficient	Does x_j matter *given the others*?
12	3. The overall F-test	Does *any* predictor matter?
10	4. Partial (nested) F-test and $F = t^2$	Does a *subset* matter?
12	5. General linear hypothesis $\mathrm{L}\boldsymbol{\beta} = \mathrm{c}$	One framework for every test
12	6. Confidence intervals and duality	What values are compatible with the data?
7	7. Wrap-up, R code, exercises	Which tool when?

By the end of this lecture you can:

1. State, in words and symbols, what the **null hypothesis** of each test says.
2. Carry out and interpret the t-test for an individual coefficient.
3. Carry out and interpret the overall F-test and the **partial (nested) F-test**.
4. Explain the identity $F_{1,\nu} = t_\nu^2$ and when each test is appropriate.
5. Test a **general linear hypothesis** $\mathrm{L}\boldsymbol{\beta} = \mathrm{c}$ with an F-statistic.
6. Build confidence intervals, interpret them correctly, and use the **duality** between intervals and tests.

Reading. Faraway (2015) ch. 3; Kutner et al. (2005) ch. 2, 6–7; Montgomery, Peck & Vining (2012) ch. 3–4; Monahan (2008) ch. 4.

Symbol	Meaning	Size / example
n	number of observations (rows of data)	scalar; $n = 32$ cars
p	number of regression coefficients, *including* the intercept	scalar; $p = 4$
y	response vector (what we want to explain)	$n \times 1$ (mpg)
X	*design matrix*: first column all 1's (intercept), then one column per predictor	$n \times p$
x_{ij}	value of predictor j for observation i	scalar
$\boldsymbol{\beta} = (\beta_0, \ldots, \beta_{p-1})^\top$	**true** coefficient vector: fixed, unknown population numbers; β_0 is the intercept	$p \times 1$
β_j	change in the mean response for a one-unit increase in x_j, *holding the other predictors fixed*	scalar
$\boldsymbol{\varepsilon}$	random errors (everything the predictors do not explain)	$n \times 1$
σ^2	error variance (unknown): $\mathrm{Var}(\varepsilon_i) = \sigma^2$	scalar
I, 0	identity matrix, zero vector (of the right size)	–

The linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N(0, \sigma^2 \mathbf{I}), \qquad \text{i.e. } y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i.$$

Symbol	Meaning
$\hat{\boldsymbol{\beta}} = (\mathsf{X}^\top\mathsf{X})^{-1}\mathsf{X}^\top\mathsf{y}$	least-squares *estimate* of $\boldsymbol{\beta}$ (a hat means "computed from data")
$\hat{\mathsf{y}} = \mathsf{X}\hat{\boldsymbol{\beta}}$	fitted values; $\hat{y}_i$ = predicted mean for observation i
$\mathsf{e} = \mathsf{y} - \hat{\mathsf{y}}$	residuals (observed minus fitted); estimates of the errors $\boldsymbol{\varepsilon}$
$\mathsf{H} = \mathsf{X}(\mathsf{X}^\top\mathsf{X})^{-1}\mathsf{X}^\top$	hat matrix: $\hat{\mathsf{y}} = \mathsf{H}\mathsf{y}$ and $\mathsf{e} = (\mathsf{I} - \mathsf{H})\mathsf{y}$
$\mathrm{SSE} = \mathsf{e}^\top\mathsf{e} = \sum_i (y_i - \hat{y}_i)^2$	*error* sum of squares: variation the model leaves unexplained
$\mathrm{SST} = \sum_i (y_i - \bar{y})^2$	*total* sum of squares: variation of y around its mean $\bar{y}$
$\mathrm{SSR} = \sum_i (\hat{y}_i - \bar{y})^2 = \mathrm{SST} - \mathrm{SSE}$	*regression* sum of squares: variation explained by the predictors
$\hat{\sigma}^2 = \mathrm{MSE} = \mathrm{SSE}/(n-p)$	estimate of σ^2 (mean squared error); $\hat{\sigma}$ = "residual standard error"
$c_{jj} = \left[(\mathsf{X}^\top\mathsf{X})^{-1}\right]_{jj}$	j-th *diagonal* entry of $(\mathsf{X}^\top\mathsf{X})^{-1}$ (rows/columns numbered $0, \ldots, p-1$)
$\widehat{\mathrm{SE}}(\hat{\beta}_j) = \hat{\sigma}\sqrt{c_{jj}}$	estimated standard error of $\hat{\beta}_j$: typical size of its sampling error
$n - p$	*degrees of freedom* (df) of the residuals: n data points minus p estimated coefficients

Convention: a symbol *with* a hat ($\hat{\beta}_j, \hat{\sigma}^2$) is a number computed from the sample; *without* a hat (β_j, σ^2) it is an unknown fixed truth.

$Z \sim N(0,1)$ standard normal.

$V \sim \chi^2_\nu$ chi-squared with ν degrees of freedom: $V = Z_1^2 + \cdots + Z_\nu^2$ for independent standard normals; $V \geq 0$, $\mathrm{E}[V] = \nu$.

$T \sim t_\nu$ Student's t with ν df:

$$T = \frac{Z}{\sqrt{V/\nu}}, \qquad Z \sim N(0,1),\ V \sim \chi^2_\nu \text{ independent.}$$

Symmetric about 0, heavier tails than $N(0,1)$; $t_\nu \to N(0,1)$ as $\nu \to \infty$.

$F \sim F_{d_1,d_2}$ F with numerator df d_1 and denominator df d_2:

$$F = \frac{U_1/d_1}{U_2/d_2}, \qquad U_1 \sim \chi^2_{d_1},\ U_2 \sim \chi^2_{d_2} \text{ independent.}$$

Takes only positive values; right-skewed; a ratio of two *average* squared quantities, so values near 1 are typical when the numerator and denominator estimate the same variance.

Quantile notation

$t_{\nu,q}$ denotes the q-quantile of t_ν: $P(T \leq t_{\nu,q}) = q$. Example: $t_{28,\,0.975} = 2.048$. Likewise $F_{d_1,d_2,\,q}$.

A1 Linearity $\mathrm{E}[\mathrm{y}] = \mathrm{X}\boldsymbol{\beta}$, with X fixed and of full column rank p (so $(\mathrm{X}^\top\mathrm{X})^{-1}$ exists).

A2 Zero-mean errors $\mathrm{E}[\boldsymbol{\varepsilon}] = 0$.

A3 Constant variance, uncorrelated $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathrm{I}$.

A4 Normality $\boldsymbol{\varepsilon} \sim N(0, \sigma^2\mathrm{I})$.

Result	Needs	Where used today
$\mathrm{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ (unbiased)	A1, A2	everything
$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathrm{X}^\top\mathrm{X})^{-1}$	A1–A3	standard errors
exact t and F distributions	A1–A4	p-values, critical values
approximate t and F (large n)	A1–A3 + CLT	robustness (Part 7)

Everything that follows is a statement about the *sampling distribution* of an estimator: how it would vary if we could repeat the study many times with the same X.

Running example: mtcars, mpg $\sim$ wt + hp + qsec

$n = 32$ cars; response $y =$ mpg; predictors $x_1 =$ wt (weight, 1000 lb), $x_2 =$ hp (gross horsepower), $x_3 =$ qsec (quarter-mile time, seconds). So $p = 4$, $n - p = 28$, $\boldsymbol{\beta} = (\beta_0, \beta_{\mathtt{wt}}, \beta_{\mathtt{hp}}, \beta_{\mathtt{qsec}})^\top$.

	Estimate $\hat\beta_j$	Std. error $\widehat{\mathrm{SE}}(\hat\beta_j)$	t_j	p-value
(Intercept)	27.6105	8.4199	3.279	0.0028
wt	−4.3588	0.7527	−5.791	3.2×10^{-6}
hp	−0.0178	0.0150	−1.190	0.2442
qsec	0.5108	0.4392	1.163	0.2546

Residual standard error $\hat\sigma = 2.578$ on 28 df ($\hat\sigma^2 = 6.645$); $R^2 = 0.8348$, adjusted $R^2 = 0.8171$; overall $F = 47.15$ on $(3, 28)$ df, $p = 4.5 \times 10^{-11}$.

Today's job

Explain *where each number in this table comes from*, what hypothesis each p-value refers to, and how to answer richer questions (several coefficients at once, equality of coefficients, intervals).

Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + \underbrace{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top}_{=:\ \mathbf{A}\ \text{(a fixed matrix)}}\ \boldsymbol{\varepsilon}.$$

- **Mean:** $\mathrm{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + \mathbf{A}\,\mathrm{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}$. The estimator is **unbiased**.
- **Variance:** $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \mathbf{A}(\sigma^2\mathbf{I})\mathbf{A}^\top = \sigma^2\mathbf{A}\mathbf{A}^\top = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$, since $\mathbf{A}\mathbf{A}^\top = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = (\mathbf{X}^\top\mathbf{X})^{-1}$.
- **Shape:** $\hat{\boldsymbol{\beta}}$ is a linear function of the normal vector $\boldsymbol{\varepsilon}$, hence normal.

$$\hat{\boldsymbol{\beta}} \sim N_p\big(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\big) \quad\Longrightarrow\quad \hat{\beta}_j \sim N\big(\beta_j,\ \sigma^2 c_{jj}\big), \qquad c_{jj} = \big[(\mathbf{X}^\top\mathbf{X})^{-1}\big]_{jj}.$$

Notation. $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: p-variate normal with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathrm{E}[\cdot]$: expectation. $\mathrm{Var}(\cdot)$: covariance matrix. Off-diagonal entries of $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ are the covariances between different $\hat{\beta}_j$ (needed later for $\mathbf{L}\boldsymbol{\beta}$).

Residuals are a projection of the errors. Since $(\mathrm{I}-\mathrm{H})\mathrm{X}=0$,

$$\mathrm{e}=(\mathrm{I}-\mathrm{H})\mathrm{y}=(\mathrm{I}-\mathrm{H})\boldsymbol{\varepsilon}, \qquad \mathrm{SSE}=\mathrm{e}^{\top}\mathrm{e}=\boldsymbol{\varepsilon}^{\top}(\mathrm{I}-\mathrm{H})\boldsymbol{\varepsilon}.$$

$\mathrm{I}-\mathrm{H}$ is symmetric and idempotent with rank $n-p$. A standard fact: if $\boldsymbol{\varepsilon}\sim N(0,\sigma^2\mathrm{I})$ and M is symmetric idempotent of rank r, then $\boldsymbol{\varepsilon}^{\top}\mathrm{M}\boldsymbol{\varepsilon}/\sigma^2\sim\chi^2_r$. Hence

$$\boxed{\frac{(n-p)\,\hat{\sigma}^2}{\sigma^2}=\frac{\mathrm{SSE}}{\sigma^2}\sim\chi^2_{n-p}} \qquad (\text{so } \mathrm{E}[\hat{\sigma}^2]=\sigma^2).$$

Independence. $\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}=(\mathrm{X}^{\top}\mathrm{X})^{-1}\mathrm{X}^{\top}\mathrm{H}\boldsymbol{\varepsilon}$ depends only on $\mathrm{H}\boldsymbol{\varepsilon}$, while $\mathrm{e}=(\mathrm{I}-\mathrm{H})\boldsymbol{\varepsilon}$. Their covariance is

$$\mathrm{Cov}\big(\mathrm{H}\boldsymbol{\varepsilon},(\mathrm{I}-\mathrm{H})\boldsymbol{\varepsilon}\big)=\sigma^2\mathrm{H}(\mathrm{I}-\mathrm{H})=0,$$

and jointly normal vectors with zero covariance are independent. So $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are **independent**.

Why $n-p$? The n residuals satisfy p linear constraints ($\mathrm{X}^{\top}\mathrm{e}=0$), leaving $n-p$ free directions: the degrees of freedom.

We have two independent ingredients:

$$Z_j = \frac{\hat\beta_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1), \qquad V = \frac{(n-p)\hat\sigma^2}{\sigma^2} \sim \chi^2_{n-p}.$$

Z_j cannot be computed (it contains the unknown σ). Form the ratio in the definition of t_ν with $\nu = n - p$:

$$T_j = \frac{Z_j}{\sqrt{V/(n-p)}} = \frac{(\hat\beta_j - \beta_j)/(\sigma\sqrt{c_{jj}})}{\hat\sigma/\sigma} = \frac{\hat\beta_j - \beta_j}{\hat\sigma\sqrt{c_{jj}}} = \frac{\hat\beta_j - \beta_j}{\widehat{\mathrm{SE}}(\hat\beta_j)} \sim t_{n-p}.$$

- The unknown σ *cancels*: the distribution of T_j does not depend on any unknown parameter. Such a quantity is called a **pivot**.
- Replacing σ by $\hat\sigma$ is what turns the normal into a t (extra uncertainty $\Rightarrow$ heavier tails).

`wt` coefficient

$c_{\mathrm{wt,wt}} = 0.08526$, $\hat\sigma^2 = 6.645$: $\widehat{\mathrm{SE}}(\hat\beta_{\mathrm{wt}}) = \sqrt{6.645 \times 0.08526} = 0.7527$.

Hypotheses (two-sided)

$$H_0 : \beta_j = \beta_j^0 \qquad \text{versus} \qquad H_1 : \beta_j \neq \beta_j^0.$$

- H_0 ("null hypothesis"): the claim we *test*, assumed true until the data contradict it; H_1 ("alternative"): what we conclude if we reject H_0.
- β_j (no hat): the **true** coefficient of x_j, a fixed unknown number; $j \in \{0, 1, \ldots, p-1\}$.
- β_j^0: the specific number we **choose before seeing the data**. The superscript 0 is a label ("value under H_0"), *not* a power and *not* an estimate.

In words

Recall β_j = change in mean response per unit increase of x_j, other predictors held fixed. So $H_0 : \beta_j = \beta_j^0$ says "that change is exactly β_j^0". The default $\beta_j^0 = 0$ says: **given the other predictors in the model, x_j has no linear effect on the mean response.**

`mtcars`

$H_0 : \beta_{\texttt{wt}} = 0$: "for cars with the same `hp` and `qsec`, weight does not change average `mpg`." (A null like $\beta_{\texttt{wt}} = -5$ asks if 1000 lb costs exactly 5 mpg.)

$$t_j = \frac{\overbrace{\hat{\beta}_j}^{\text{estimate}} - \overbrace{\beta_j^0}^{\text{hypothesised value}}}{\underbrace{\widehat{\mathrm{SE}}(\hat{\beta}_j)}_{\text{estimated standard error}}} \sim t_{n-p} \quad \text{when } H_0 \text{ is true.}$$

- **Numerator** = *signal*: how far the data-based estimate is from what H_0 claims.
- **Denominator** = *noise*: how far $\hat{\beta}_j$ typically strays from β_j by chance alone.
- So t_j counts *how many standard errors* the estimate lies from the hypothesised value. Its sign gives the direction; $|t_j|$ the strength of evidence against H_0.
- Reference distribution t_{n-p}: it is exact because of Part 1 (pivot). Under H_0, t_j is typically between about -2 and 2.

wt, $\beta_{\text{wt}}^0 = 0$

$$t_{\text{wt}} = \frac{-4.3588 - 0}{0.7527} = -5.791 \quad \text{on } n - p = 28 \text{ df.}$$

The estimate is almost six standard errors below zero: very surprising if β_{wt} were truly 0.

Choose the significance level α (commonly 0.05) *in advance*: the probability of rejecting a true H_0 (**Type I error**).

Critical-value rule

Reject H_0 iff $|t_j| > t_{n-p,\,1-\alpha/2}$.

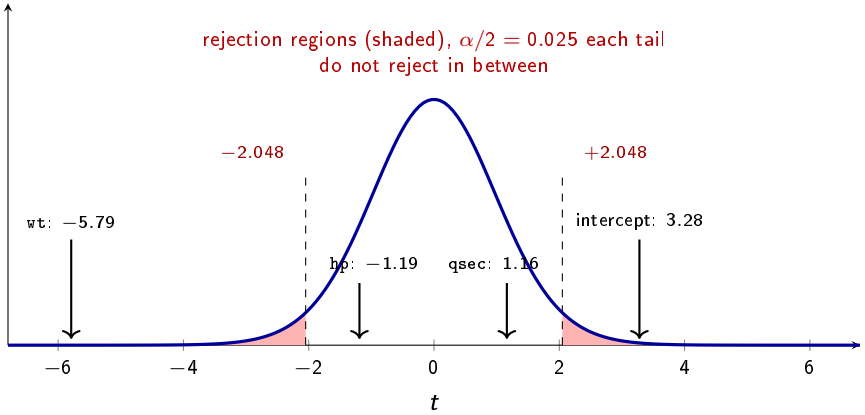
Here $t_{28,0.975} = 2.048$.

p-value rule

$$p = 2\,P\big(T_{n-p} \geq |t_j^{\text{obs}}|\big)$$

Reject H_0 iff $p < \alpha$.

- T_{n-p} is a *random variable* with the t_{n-p} distribution; t_j^{obs} is the *number* we computed. The factor 2 counts both tails (two-sided).
- The two rules are equivalent: $|t_j| > t_c \iff p < \alpha$.
- **One-sided** alternatives, e.g. $H_1 : \beta_j > \beta_j^0$: reject for large positive t_j; $p = P(T_{n-p} \geq t_j^{\text{obs}})$, critical value $t_{n-p,1-\alpha}$.
- **Type II error** = failing to reject a false H_0; **power** $= 1 - P(\text{Type II})$ grows with $|\beta_j - \beta_j^0|$ and n, and shrinks with σ.



wt and the intercept fall in the rejection region; hp and qsec do not.

Test $H_0 : \beta_j = 0$ vs $H_1 : \beta_j \neq 0$ at $\alpha = 0.05$, critical value $t_{28,0.975} = 2.048$.

H_0	$\hat{\beta}_j$	$\widehat{\mathrm{SE}}(\hat{\beta}_j)$	$t_j = \hat{\beta}_j / \widehat{\mathrm{SE}}$	p-value	Decision
$\beta_0 = 0$	27.6105	8.4199	3.279	0.0028	reject
$\beta_{\mathrm{wt}} = 0$	−4.3588	0.7527	−5.791	3.2×10^{-6}	reject
$\beta_{\mathrm{hp}} = 0$	−0.0178	0.0150	−1.190	0.2442	do not reject
$\beta_{\mathrm{qsec}} = 0$	0.5108	0.4392	1.163	0.2546	do not reject

Reading the result. Holding `hp` and `qsec` fixed, each extra 1000 lb is associated with about 4.36 mpg *lower* mean mpg, and this is clearly not zero. For `hp` and `qsec` the data are *compatible with* no partial effect.

Careful

"Do not reject H_0" is **not** "H_0 is true": it means the data do not provide enough evidence against it (the confidence interval, Part 6, shows how large the effect could still be).

The t-test asks: *does x_j help beyond the other predictors already in the model?* Change the model and the answer can change.

	slope alone (simple regression)		slope in `wt+hp+qsec`	
	$\hat{\beta}$	p	$\hat{\beta}$	p
`wt`	-5.344	1.3×10^{-10}	-4.359	3.2×10^{-6}
`hp`	-0.0682	1.8×10^{-7}	-0.0178	0.244
`qsec`	1.412	0.017	0.511	0.255

`hp` and `qsec` are strongly correlated ($r = -0.71$): once one is in the model, the other adds little.

The p-value is *not*...

- the probability that H_0 is true;
- the probability the result will replicate;
- a measure of effect size.

It *is*...

- the probability, *if H_0 were true*, of a t statistic at least as extreme as the one observed.

Hypotheses

$$H_0: \ \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0 \qquad \text{versus} \qquad H_1: \ \beta_j \neq 0 \text{ for at least one } j \in \{1, \ldots, p-1\}.$$

- The intercept β_0 is *not* part of H_0; $\beta_1, \ldots, \beta_{p-1}$ are the $p-1$ *slopes*.
- Under H_0 the model collapses to $y_i = \beta_0 + \varepsilon_i$ (the **intercept-only model**); its least-squares fit is $\hat{\beta}_0 = \bar{y}$.
- H_1 does *not* say which slopes are non-zero or how many.

In words

H_0: **none of the predictors is linearly related to the mean response**; the mean of y is the same constant whatever the x's are.

`mtcars`

$H_0: \beta_{\text{wt}} = \beta_{\text{hp}} = \beta_{\text{qsec}} = 0$: "weight, horsepower and quarter-mile time *all together* tell us nothing about mean mpg". Here $p - 1 = 3$ restrictions.

When the model contains an intercept,

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST (total)}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR (explained)}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE (unexplained)}}, \qquad n - 1 = (p - 1) + (n - p).$$

Each sum of squares divided by its degrees of freedom is a **mean square**: $\mathrm{MSR} = \mathrm{SSR}/(p-1)$, $\mathrm{MSE} = \mathrm{SSE}/(n-p) = \hat{\sigma}^2$.

Source	df	SS	MS	F
Regression	$p-1$	SSR	$\mathrm{MSR} = \dfrac{\mathrm{SSR}}{p-1}$	$\dfrac{\mathrm{MSR}}{\mathrm{MSE}}$
Residual	$n-p$	SSE	$\mathrm{MSE} = \dfrac{\mathrm{SSE}}{n-p}$	
Total	$n-1$	SST		

Idea. SST is the error sum of squares of the intercept-only model, i.e. of H_0. $\mathrm{SSR} = \mathrm{SST} - \mathrm{SSE}$ is how much the predictors *reduce* the error. The F-test asks whether that reduction is large compared with the noise level MSE.

Under H_0 (and A1–A4), $\mathrm{SSR}/\sigma^2 \sim \chi^2_{p-1}$, $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$, and the two are independent. Therefore

$$F = \frac{\mathrm{SSR}/(p-1)}{\mathrm{SSE}/(n-p)} = \frac{(\mathrm{SSR}/\sigma^2)/(p-1)}{(\mathrm{SSE}/\sigma^2)/(n-p)} = \frac{\mathrm{MSR}}{\mathrm{MSE}} \sim F_{p-1,\,n-p} \qquad (\sigma^2 \text{ cancels}).$$

Why a large F signals H_1. Always $\mathrm{E}[\mathrm{MSE}] = \sigma^2$. But

$$\mathrm{E}[\mathrm{MSR}] = \sigma^2 + \frac{1}{p-1}\,\boldsymbol{\beta}_S^\top \mathsf{X}_c^\top \mathsf{X}_c\,\boldsymbol{\beta}_S \;\geq\; \sigma^2,$$

with equality iff $\boldsymbol{\beta}_S = 0$ (i.e. iff H_0). So $F \approx 1$ under H_0 and $F > 1$ under H_1.

- $\boldsymbol{\beta}_S = (\beta_1, \ldots, \beta_{p-1})^\top$: the vector of slopes.
- X_c: the $n \times (p-1)$ matrix of predictors with each column centred (column mean subtracted).

Decision

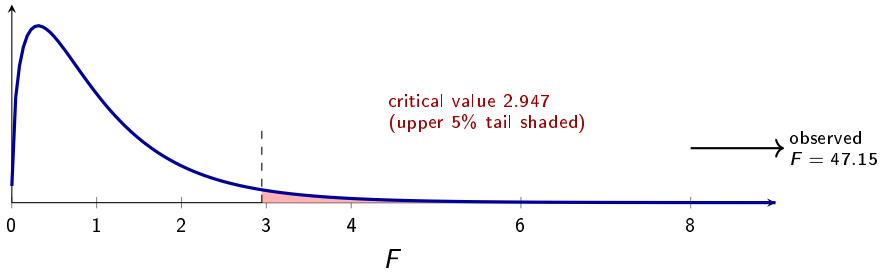
Reject H_0 iff $F > F_{p-1,n-p,1-\alpha}$, equivalently $p = P\big(F_{p-1,n-p} \geq F^{\mathrm{obs}}\big) < \alpha$. Only the *upper* tail counts.

Source	df	SS	MS	F
Regression	3	939.99	313.33	47.15
Residual	28	186.06	6.645	
Total	31	1126.05		

- $F = 313.33/6.645 = 47.15$ on $(3, 28)$ df. Critical value $F_{3,28,0.95} = 2.947$; $p = 4.5 \times 10^{-11}$. **Reject H_0.**
- **Link to R^2.** With $R^2 = \mathrm{SSR}/\mathrm{SST} = 939.99/1126.05 = 0.8348$,

$$F = \frac{R^2/(p-1)}{(1-R^2)/(n-p)} = \frac{0.8348/3}{0.1652/28} = 47.15.$$

  A larger R^2 (for fixed n, p) gives a larger F; but the same R^2 is far more convincing with $n = 1000$ than with $n = 6$.
- This is the F reported at the bottom of `summary(lm(...))`.



Significant $F \neq$ "we know which predictors matter"

F says the predictors *jointly* explain variation. It is silent about *which* ones. Conversely (collinearity paradox, Lecture 10) F can be highly significant while *every* individual t is not.

We want to test whether a *subset* of q coefficients (not necessarily all slopes) are zero. Split the predictors into those we keep (A) and those under test (B):

$$\text{Full: } \mathrm{y} = \mathrm{X}_A\boldsymbol{\beta}_A + \mathrm{X}_B\boldsymbol{\beta}_B + \boldsymbol{\varepsilon}, \qquad H_0 : \ \boldsymbol{\beta}_B = 0 \ \ (q \text{ coefficients}), \qquad \text{Reduced: } \mathrm{y} = \mathrm{X}_A\boldsymbol{\beta}_A + \boldsymbol{\varepsilon}.$$

- q = number of coefficients set to zero = number of restrictions = (parameters in full) − (parameters in reduced).
- $\mathrm{SSE}_{\text{full}}$, $\mathrm{SSE}_{\text{red}}$: error sums of squares of the two fitted models. Always $\mathrm{SSE}_{\text{red}} \geq \mathrm{SSE}_{\text{full}}$ (fewer parameters cannot fit better).
- The full model has p parameters, so its residual df is $n - p$.

The statistic

$$F = \frac{(\mathrm{SSE}_{\text{red}} - \mathrm{SSE}_{\text{full}})/q}{\mathrm{SSE}_{\text{full}}/(n-p)} \ \sim \ F_{q,\,n-p} \quad \text{under } H_0.$$

Reading it. Numerator: *average loss of fit per dropped coefficient*. Denominator: the noise level $\hat{\sigma}^2$ of the full model. Under H_0 the ratio is about 1. Equivalent form: $F = \dfrac{(R^2_{\text{full}} - R^2_{\text{red}})/q}{(1 - R^2_{\text{full}})/(n-p)}$. In R: `anova(reduced, full)`.

$H_0 : \beta_{\texttt{qsec}} = 0$ (so $q = 1$). Reduced model: `mpg~wt+hp`; full model: `mpg~wt+hp+qsec`.

Model	Residual df	SSE	
Reduced (`wt+hp`)	29	195.048	
Full (`wt+hp+qsec`)	28	186.059	$\hat{\sigma}^2_{\text{full}} = 186.059/28 = 6.645$
Difference	$q = 1$	8.988	

$$F = \frac{(195.048 - 186.059)/1}{186.059/28} = \frac{8.988}{6.645} = 1.353 \quad \text{on } (1, 28) \text{ df}, \qquad p = 0.2546.$$

Critical value $F_{1,28,0.95} = 4.196$. Since $1.353 < 4.196$: **do not reject**: `qsec` does not add significantly once `wt` and `hp` are in the model.

Cross-check with the *t*-test

From Part 2: $t_{\texttt{qsec}} = 1.163$ and $t^2_{\texttt{qsec}} = 1.163^2 = 1.353$; its p-value was also 0.2546. The two tests agree *exactly*: next slide.

(a) Distribution. If $T = Z/\sqrt{V/\nu} \sim t_\nu$ then

$$T^2 = \frac{Z^2}{V/\nu} = \frac{Z^2/1}{V/\nu}, \qquad Z^2 \sim \chi^2_1 \text{ indep. of } V \sim \chi^2_\nu \implies T^2 \sim F_{1,\nu}.$$

(b) Algebra for the same data. For $H_0 : \beta_j = 0$ one finds (proved with the general linear hypothesis, Part 5)

$$\mathrm{SSE}_{\text{red}} - \mathrm{SSE}_{\text{full}} = \frac{\hat\beta_j^2}{c_{jj}} \quad \implies \quad F = \frac{\hat\beta_j^2/c_{jj}}{\hat\sigma^2} = \Big(\frac{\hat\beta_j}{\hat\sigma\sqrt{c_{jj}}}\Big)^2 = t_j^2.$$

Numerical check (qsec): $\hat\beta^2/c = 0.5108^2/0.02903 = 8.988 = \mathrm{SSE}_{\text{red}} - \mathrm{SSE}_{\text{full}}$ ✓ **(c) Consequences.**

- Critical values match: $t^2_{28,0.975} = 2.048^2 = 4.196 = F_{1,28,0.95}$; the two-sided t p-value equals the F p-value.
- With $q = 1$ the t-test and the F-test are the *same test*. The F-test additionally handles $q \geq 2$ restrictions at once; the t-test can also be one-sided (the sign is lost when squaring).

$H_0 : \beta_{\texttt{hp}} = \beta_{\texttt{qsec}} = 0$ (so $q = 2$). Reduced model: $\texttt{mpg} \sim \texttt{wt}$, $\text{SSE}_{\text{red}} = 278.322$ (df 30); full: $\text{SSE}_{\text{full}} = 186.059$ (df 28).

$$F = \frac{(278.322 - 186.059)/2}{186.059/28} = \frac{46.131}{6.645} = 6.942 \quad \text{on } (2, 28), \quad p = 0.0036, \quad F_{2,28,0.95} = 3.340.$$

Reject H_0 – yet the individual tests gave $p = 0.244$ (hp) and $p = 0.255$ (qsec). Why?

- hp and qsec are correlated ($r = -0.71$) and each partly substitutes for the other. Given the other is in the model, each adds little (t-tests), but *jointly* they add a lot.
- Each t-test is conditional on the other variable being present; neither accounts for the correlation between $\hat{\beta}_{\texttt{hp}}$ and $\hat{\beta}_{\texttt{qsec}}$, which the F-test uses.
- Two separate tests at $\alpha = 0.05$ also do not control the overall error rate (multiple testing).

Take-away

To ask whether a *group* of predictors matters, use the partial F-test, not a collection of t-tests.

Hypothesis

$$H_0: \ \mathrm{L}\boldsymbol{\beta} = \mathrm{c} \qquad \text{versus} \qquad H_1: \ \mathrm{L}\boldsymbol{\beta} \neq \mathrm{c}.$$

- L: known $q \times p$ matrix of **rank** q; each *row* encodes one linear restriction on $\boldsymbol{\beta}$.
- c: known $q \times 1$ vector of target values (usually 0).
- q: number of independent restrictions (the numerator df).

Here $\boldsymbol{\beta} = (\beta_0, \beta_{\mathsf{wt}}, \beta_{\mathsf{hp}}, \beta_{\mathsf{qsec}})^\top$:

Hypothesis in words	H_0	L (rows)	c	q
single coefficient	$\beta_{\mathsf{hp}} = 0$	$(0,0,1,0)$	0	1
specific value	$\beta_{\mathsf{wt}} = -5$	$(0,1,0,0)$	-5	1
all slopes (overall F)	$\beta_{\mathsf{wt}} = \beta_{\mathsf{hp}} = \beta_{\mathsf{qsec}} = 0$	$(0,1,0,0); (0,0,1,0); (0,0,0,1)$	0	3
a subset	$\beta_{\mathsf{hp}} = \beta_{\mathsf{qsec}} = 0$	$(0,0,1,0); (0,0,0,1)$	0	2
equal coefficients	$\beta_{\mathsf{wt}} - \beta_{\mathsf{qsec}} = 0$	$(0,1,0,-1)$	0	1

Step 1 (distribution of the discrepancy). Let $\mathrm{d} = \mathrm{L}\hat{\boldsymbol{\beta}} - \mathrm{c}$ ($q \times 1$). From $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathrm{X}^\top\mathrm{X})^{-1})$:

$$\mathrm{d} \sim N_q\big(\mathrm{L}\boldsymbol{\beta} - \mathrm{c},\ \sigma^2\mathrm{M}\big), \qquad \mathrm{M} = \mathrm{L}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{L}^\top \ (q \times q,\ \text{invertible since } \operatorname{rank}\mathrm{L} = q).$$

Step 2 (quadratic form). Under H_0 the mean is 0. If $\mathrm{d} \sim N_q(0, \boldsymbol{\Sigma})$ then $\mathrm{d}^\top\boldsymbol{\Sigma}^{-1}\mathrm{d} \sim \chi^2_q$, so

$$\frac{\mathrm{d}^\top\mathrm{M}^{-1}\mathrm{d}}{\sigma^2} \sim \chi^2_q.$$

Step 3 (estimate σ^2, form the ratio). d is a function of $\hat{\boldsymbol{\beta}}$, hence independent of $\hat{\sigma}^2$, and $(n-p)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p}$. Dividing each chi-squared by its df:

$$\boxed{F = \frac{(\mathrm{L}\hat{\boldsymbol{\beta}} - \mathrm{c})^\top\big[\mathrm{L}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{L}^\top\big]^{-1}(\mathrm{L}\hat{\boldsymbol{\beta}} - \mathrm{c})}{q\,\hat{\sigma}^2} \ \sim\ F_{q,\,n-p}}$$

Reading it. $\mathrm{d}^\top\mathrm{M}^{-1}\mathrm{d}$ is the squared distance of $\mathrm{L}\hat{\boldsymbol{\beta}}$ from c measured in units of its own (correlated) uncertainty; dividing by q gives a per-restriction average; dividing by $\hat{\sigma}^2$ makes it scale-free. It equals $\mathrm{SSE}_{\mathrm{red}} - \mathrm{SSE}_{\mathrm{full}}$ for the restricted fit.

Write the null as $\beta_{\text{wt}} - \beta_{\text{qsec}} = 0$: $\mathrm{L} = (0,\ 1,\ 0,\ -1)$, $\mathrm{c} = 0$, $q = 1$.

1. **Discrepancy** (a scalar): $d = \mathrm{L}\hat{\boldsymbol{\beta}} - 0 = \hat{\beta}_{\text{wt}} - \hat{\beta}_{\text{qsec}} = -4.3588 - 0.5108 = -4.8696$.
2. **Its variance factor**: $M = \mathrm{L}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{L}^\top = c_{\text{wt,wt}} + c_{\text{qsec,qsec}} - 2c_{\text{wt,qsec}}$ (variance of a difference = sum of variances minus twice the covariance), with entries of $(\mathrm{X}^\top\mathrm{X})^{-1}$:

$$M = 0.08526 + 0.02903 - 2(-0.02733) = 0.16896.$$

3. **Statistic**:

$$F = \frac{d^2}{q\, M\, \hat{\sigma}^2} = \frac{(-4.8696)^2}{1 \times 0.16896 \times 6.645} = \frac{23.713}{1.1228} = 21.12 \quad \text{on } (1, 28), \quad p = 8.4 \times 10^{-5}.$$

Reject H_0: the data contradict equal `wt` and `qsec` coefficients.

Statistical vs. substantive sense

`wt` (1000 lb) and `qsec` (seconds) have different units, so "equal coefficients" is rarely a meaningful scientific claim; we use it here because it illustrates the mechanics with $q = 1$. Equality tests are sensible for variables on the same scale.

If $\beta_{\texttt{wt}} = \beta_{\texttt{qsec}} =: \gamma$, then

$$\beta_{\texttt{wt}}\,\texttt{wt} + \beta_{\texttt{qsec}}\,\texttt{qsec} = \gamma\,(\texttt{wt} + \texttt{qsec}),$$

so the *reduced model* is $\texttt{mpg} = \beta_0 + \gamma\,(\texttt{wt} + \texttt{qsec}) + \beta_{\texttt{hp}}\,\texttt{hp} + \varepsilon$: in R, `lm(mpg ~ I(wt + qsec) + hp)`.

	Route A: quadratic form	Route B: nested models
Needs	full model fit only	fit full *and* reduced
Statistic	$\dfrac{\mathrm{d}^\top \mathrm{M}^{-1}\mathrm{d}}{q\hat\sigma^2} = \dfrac{(-4.8696)^2/0.16896}{6.645}$	$\dfrac{(\mathrm{SSE}_{\text{red}} - \mathrm{SSE}_{\text{full}})/q}{\hat\sigma^2} = \dfrac{(326.407 - 186.059)}{6.645}$
Numerator	140.35	140.35
F	21.12	21.12

Identical. In general $\mathrm{SSE}_{\text{red}} - \mathrm{SSE}_{\text{full}} = (\mathrm{L}\hat{\boldsymbol\beta} - \mathrm{c})^\top[\mathrm{L}(\mathrm{X}^\top\mathrm{X})^{-1}\mathrm{L}^\top]^{-1}(\mathrm{L}\hat{\boldsymbol\beta} - \mathrm{c})$. Route A is convenient when many hypotheses are tested on one fit (software: `car::linearHypothesis`); Route B when the restricted model is easy to write down.

Pivot (Part 1): for the *true* β_j, $T_j = \dfrac{\hat\beta_j - \beta_j}{\widehat{\mathrm{SE}}(\hat\beta_j)} \sim t_{n-p}$. With $t_c = t_{n-p,\,1-\alpha/2}$:

$$P\big(-t_c \le T_j \le t_c\big) = 1-\alpha \iff P\big(\hat\beta_j - t_c\,\widehat{\mathrm{SE}}(\hat\beta_j) \;\le\; \beta_j \;\le\; \hat\beta_j + t_c\,\widehat{\mathrm{SE}}(\hat\beta_j)\big) = 1-\alpha.$$

$100(1-\alpha)\%$ confidence interval

$$\hat\beta_j \;\pm\; \underbrace{t_{n-p,\,1-\alpha/2}\,\widehat{\mathrm{SE}}(\hat\beta_j)}_{\text{margin of error}}.$$

Interpretation.

- Before the data are collected, the *endpoints are random* and β_j is fixed: 95% of intervals built this way (over repeated samples) contain β_j.
- For one *computed* interval, β_j is either inside or not. Say "we are 95% confident" (a statement about the *method*), not "β_j lies in this interval with probability 0.95".
- Margin of error grows with $\hat\sigma$, with the confidence level, and with collinearity (large c_{jj}); it shrinks with n.

Critical values on 28 df: $t_{28,0.975} = 2.048$ (95%), $t_{28,0.995} = 2.763$ (99%).

	$\hat{\beta}_j$	$\widehat{\mathrm{SE}}$	95% CI	99% CI
(Intercept)	27.611	8.420	(10.363, 44.858)	(4.344, 50.877)
`wt`	−4.359	0.753	(−5.901, −2.817)	(−6.439, −2.279)
`hp`	−0.0178	0.0150	(−0.0485, 0.0129)	(−0.0592, 0.0236)
`qsec`	0.511	0.439	(−0.389, 1.411)	(−0.703, 1.725)

Example: `wt` : $-4.3588 \pm 2.048 \times 0.7527 = -4.3588 \pm 1.5418 = (-5.901, -2.817)$.

- **Effect size**: with `hp`,`qsec` fixed, an extra 1000 lb is associated with between 2.8 and 5.9 mpg lower mean mpg (95%).
- **99% intervals are wider** because $2.763 > 2.048$: more confidence costs precision. The matching test at $\alpha = 0.01$ rejects less easily.

Claim

The two-sided level-α t-test of $H_0 : \beta_j = \beta_j^0$ rejects **if and only if** β_j^0 lies *outside* the $100(1-\alpha)\%$ confidence interval.

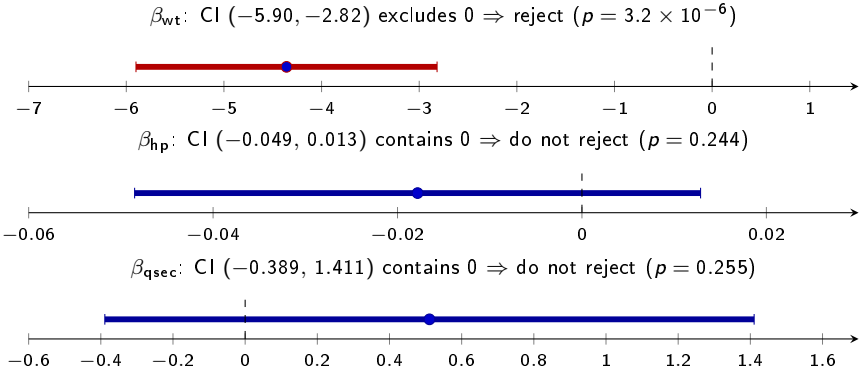
Proof. With $t_c = t_{n-p,1-\alpha/2}$:

$$\beta_j^0 \in \text{CI} \iff \left|\hat{\beta}_j - \beta_j^0\right| \le t_c \,\widehat{\text{SE}}(\hat{\beta}_j) \iff \left|\frac{\hat{\beta}_j - \beta_j^0}{\widehat{\text{SE}}(\hat{\beta}_j)}\right| = |t_j| \le t_c \iff \text{do not reject.} \qquad \square$$

Consequences.

- The CI is the set of *all* null values β_j^0 that the data do not reject at level α: the *compatible* values.
- One interval answers infinitely many tests at once. `hp`: the 95% CI $(-0.0485, 0.0129)$ contains 0 ($p = 0.244$), but also -0.02, -0.04, $+0.01\ldots$: so we cannot call the effect "zero", only "not shown to differ from zero".
- Check on `mtcars`: for all four coefficients, "CI excludes 0" $\Leftrightarrow$ "$p < 0.05$".

A CI reports effect size and precision, not just reject/retain: the antidote to confusing *significance* with *importance*.



Dashed line: the null value $\beta_j^0 = 0$. Dot: $\hat{\beta}_j$. Bar: 95% CI (each panel has its own horizontal scale).

Let $x_0 = (1, x_{01}, \ldots, x_{0,p-1})^\top$ be a fixed predictor vector. The **mean response** there is $\mu_0 = x_0^\top \beta$, estimated by $\hat{\mu}_0 = x_0^\top \hat{\beta}$ with $\mathrm{Var}(\hat{\mu}_0) = \sigma^2 h_{00}, \quad h_{00} = x_0^\top (X^\top X)^{-1} x_0$.

CI for the mean response μ_0

$$\hat{\mu}_0 \pm t_{n-p,1-\alpha/2}\, \hat{\sigma} \sqrt{h_{00}}$$

Prediction interval for a new y_0

$$\hat{\mu}_0 \pm t_{n-p,1-\alpha/2}\, \hat{\sigma} \sqrt{1 + h_{00}}$$

The "+1" is the irreducible variance σ^2 of the new error ε_0; developed in Lecture 5.

$x_0 = (1,\ 3,\ 150,\ 18)^\top$: a 3000 lb car, 150 hp, quarter-mile 18 s

$\hat{\mu}_0 = 27.6105 - 4.3588(3) - 0.017822(150) + 0.510834(18) = 21.056$ mpg; $h_{00} = 0.04078$.

95% CI for mean mpg: $21.056 \pm 2.048 \times \sqrt{6.645 \times 0.04078} = 21.056 \pm 1.066 = (19.99,\ 22.12)$.

95% prediction interval for one such car:
$21.056 \pm 2.048 \times \sqrt{6.645 \times 1.04078} = 21.056 \pm 5.386 = (15.67,\ 26.44)$.

We know the *average* mpg of such cars far better than we can predict any *single* car.

Exact t and F distributions rely on A4 (normal errors). If A1–A3 hold but errors are non-normal:

- With large n (and no dominating high-leverage points), $\hat{\boldsymbol{\beta}}$ is approximately normal by the CLT and $\hat{\sigma} \to \sigma$. Then $t_{n-p} \approx N(0,1)$ and the tests and intervals are **approximately valid** (Lecture 3). For 28 df: $t_c = 2.048$ vs. $z_c = 1.960$.
- With **small** n and heavy tails/skewness, the actual error rates can differ noticeably from the nominal α.

What breaks the procedures even for large n:

- **Non-constant variance** (A3 fails): the formula $\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$ is wrong, so standard errors, t, F and CIs are all off $\Rightarrow$ robust (sandwich) standard errors.
- **Dependence** between errors, **outliers**/influential points, an incorrect mean model (A1).

What to do

Check residuals (normal Q–Q plot, residuals vs. fitted values); use heteroskedasticity-robust SEs or the **bootstrap** when A3/A4 are doubtful. **Next lecture**: inference without normality.

Summary: which tool for which question?

Question	Null hypothesis	Statistic	Reference	R
One coefficient (partial effect)	$\beta_j = \beta_j^0$	t_j	t_{n-p}	`summary(fit)`
Does *any* slope matter?	$\beta_1 = \cdots = \beta_{p-1} = 0$	overall F	$F_{p-1,n-p}$	`summary(fit)`
Does a *subset* of q matter?	those q coefficients $= 0$	partial F	$F_{q,n-p}$	`anova(red, full)`
Any linear restriction	$\mathrm{L}\boldsymbol{\beta} = \mathrm{c}$	F (quadratic form)	$F_{q,n-p}$	`linearHypothesis`
Plausible values of β_j	–	$\hat{\beta}_j \pm t_c \widehat{\mathrm{SE}}$	t_{n-p}	`confint(fit)`

- t is a *partial* question; overall F significant $\neq$ which predictors matter; for $q = 1$, $F = t^2$.
- $\mathrm{L}\boldsymbol{\beta} = \mathrm{c}$ unifies all tests and equals the nested-model comparison.
- CIs are dual to tests and report effect size: prefer them.

Next

Lecture 5: inference *without* normality (the bootstrap) and prediction intervals.

Reproducing everything in R

```r
fit <- lm(mpg ~ wt + hp + qsec, data = mtcars)
summary(fit)                  # estimates, SE, t, p; sigma-hat; R^2; overall F
confint(fit, level = 0.95)    # 95% CIs  (use level = 0.99 for 99%)

red <- lm(mpg ~ wt + hp, data = mtcars)
anova(red, fit)               # partial F for qsec: F = 1.353 on (1, 28)

red2 <- lm(mpg ~ wt, data = mtcars)
anova(red2, fit)              # joint test hp = qsec = 0: F = 6.942 on (2, 28)

library(car)
linearHypothesis(fit, "wt = qsec")             # F = 21.12 on (1, 28)
linearHypothesis(fit, c("hp = 0", "qsec = 0")) # same as anova(red2, fit)

fit0 <- lm(mpg ~ I(wt + qsec) + hp, data = mtcars)
anova(fit0, fit)              # nested-model route: F = 21.12

X <- model.matrix(fit); XtXi <- solve(t(X) %*% X)   # the matrix (X'X)^{-1}
L <- matrix(c(0, 1, 0, -1), nrow = 1); d <- L %*% coef(fit)
drop(t(d) %*% solve(L %*% XtXi %*% t(L)) %*% d) / summary(fit)$sigma^2

predict(fit, data.frame(wt = 3, hp = 150, qsec = 18), interval = "confidence")
predict(fit, data.frame(wt = 3, hp = 150, qsec = 18), interval = "prediction")
```

Check your understanding (5 minutes)

1. The overall F is highly significant but every individual t is not. Contradiction?
   No. F tests all slopes jointly; correlated predictors share explanatory power, so each adds little given the others (Lecture 10).
2. The 95% CI for $\beta_{\texttt{hp}}$ is $(-0.049, 0.013)$. What do you conclude about $H_0 : \beta_{\texttt{hp}} = 0$ and about $H_0 : \beta_{\texttt{hp}} = -0.02$ at $\alpha = 0.05$?
   Neither is rejected: both values lie inside the interval (duality).
3. Why do the t-tests use $n - p = 28$ degrees of freedom and not $n = 32$?
   Four coefficients were estimated; each uses up one degree of freedom of the residuals.
4. A partial F with $q = 1$ equals 1.353. What is $|t|$ for that coefficient?
   $|t| = \sqrt{1.353} = 1.163$.
5. What does $H_0 : \beta_{\texttt{wt}} = 0$ say in plain language?
   Among cars with identical `hp` and `qsec`, weight has no linear effect on mean `mpg`.

Exercises

1. Verify $F = t^2$ for a different single coefficient in a model of your choice, numerically *and* by deriving it from the general-linear-hypothesis formula.
   Hint: take $\mathrm{L} = \mathrm{e}_j^\top$ (the j-th unit row vector), $\mathrm{c} = 0$, so $\mathrm{M} = c_{jj}$ and $F = \hat{\beta}_j^2 / (c_{jj}\hat{\sigma}^2)$.
2. Test the joint hypothesis that **two** coefficients are simultaneously zero with a partial F, and explain why this is not the same as two separate t-tests.
   Hint: compare with the `hp`, `qsec` example; think about correlation of the estimators.
3. Construct L to test that three group means are equal in a model with two dummy variables, and carry out the F-test.
   Hint: $y = \beta_0 + \beta_1 d_1 + \beta_2 d_2 + \varepsilon$ with group A as baseline; equal means $\iff \beta_1 = \beta_2 = 0$, so $\mathrm{L} = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, $q = 2$.
4. Show that widening the confidence level from 95% to 99% widens the interval and makes the corresponding test less likely to reject; relate this to the t critical value.
   Hint: $t_{28,0.995} = 2.763 > t_{28,0.975} = 2.048$; use duality.
5. With a large simulated sample, demonstrate that the t-test remains approximately valid under mildly non-normal errors (e.g. t_5 or skewed), illustrating the CLT argument from Lecture 3.
   Hint: simulate under H_0, record the rejection rate at $\alpha = 0.05$ for $n = 20, 50, 200$; it should approach 0.05.